PART II: PATTERNS

Topic 7: Trend — The Long-Term Direction

The Core Concept

The **trend** is the overall, long-term direction of the data — ignoring short-term zig-zags. It tells you whether a situation is generally **improving, worsening, or stable** over months or years.

Why It Matters: One Bad Month ≠ Failure

A single high-incident month often triggers political or media attention. But if the long-term trend is downward, that spike may be a temporary fluctuation in an otherwise improving situation — not evidence of failure.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

The summer spike in 2023 (June = 22, July = 25) looks alarming in isolation. But in context:

- 2021 summer peak: 42 incidents
- 2022 summer peak: 35 incidents
- 2023 summer peak: 25 incidents

The peaks are getting lower every year. The long-term trend is clearly downward, even though month-to-month, the pattern looks noisy.

💡 **The Insight:** The trend is the big picture. A single bad month does not erase years of improvement — and a single good month does not confirm that a crisis is over.